Deven Desai

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SUMMARY

Software Engineering Master's student and Stanford Hackathon winner with strong systems programming fundamentals, C++ and Python fluency, and hands-on experience building high-reliability, real-time data execution pipelines.

PROJECTS

High-Performance Real-Time Search Engine | Low-Latency Processing Engine GitHub

- Queried and analyzed high-volume structured telemetry datasets in real time by engineering a custom high-performance indexing interface as a solo developer using C++ and Java.
- Decreased search query latency by 35 percent by developing low-level memory access routines, custom indexing data structures, and cache-friendly algorithms as measured by benchmark traces.
- Optimized processing throughput by 20 percent by profiling memory allocations, tuning concurrency primitives, and eliminating execution bottlenecks.

AgenticFlow | Concurrent Distributed Systems Engine GitHub

- Automated schema migrations and data transformations across cloud systems by architecting a high-throughput distributed execution service using Go, Python, and gRPC.
- Increased service execution throughput by 40 percent by designing concurrent execution loops, lock-free communication patterns, and low-overhead microservice interfaces.
- Maintained strict operational reliability across platform execution boundaries by implementing an automated differential privacy and verification layer.

OmniTrack AI | Multimodal Edge Processing & Monitoring Platform GitHub

- Processed unstructured live streaming inputs and metadata natively in cloud infrastructure by engineering a zero-to-one full-stack data processing application using Python and React.
- Achieved sub-100ms vector retrieval latency by independently optimizing Qdrant indexing algorithms for compute-intensive, concurrent requests.
- Enhanced system stability and operational uptime by 30 percent by building continuous execution monitoring and automated testing suites.

WORK EXPERIENCE

Applied AI Intern, *Parspec, Inc., USA* **May 2026 – Dec 2026**

- Improved product-search relevance over a distributor inventory catalog, raising **recall@3 from 50% to 90%** on a **5,000-query** labeled evaluation set, by replacing keyword-only search with **hybrid** lexical and **semantic retrieval** fused via **reciprocal rank fusion**, re-scored with a **cross-encoder re-ranker**.
- Eliminated manual review of several-hundred-page construction documents for distributor bids, extracting structured product records at **88%** precision across **CSI Divisions 26 and 22**, by building a **RAG** pipeline over written specifications and schedules that grounds every extracted field in its source excerpt and auto-enriches contractor **Quantity Take-Off** line items.
- Automated lighting fixture product substitutions across 1.2M+ indexed datasheets at 85 percent match precision by engineering a retrieval pipeline that resolves free text inputs to anchor products and scores candidates via multi attribute filtering and semantic search.

Full Stack Developer, *Terminal Trends, India* **May 2024 – Dec 2024**

- Developed and optimized RESTful APIs using **Node.js** and **Express** to integrate real-time EV charging station data from multiple vendors (e.g., Tesla, government APIs), reducing server response time by **40%** and improving data reliability across the platform.
- Built interactive route-planning features with **React** and **Redux**, enabling users to locate nearby EV stations, view waiting times, pricing per kWh, and other live metrics along their journey, enhancing user convenience and engagement.

TECHNICAL SKILLS

Languages: C++, Python, Java, Go, TypeScript, C, SQL, Shell Scripting
Systems & IoT Engineering: Real-Time Systems, Embedded Systems Concepts, Concurrency & Multithreading, Memory Management
Full Stack & Cloud Infrastructure: React, Node.js, RESTful APIs, gRPC Microservices, Docker, CI/CD, Cloud Platforms (AWS
Tools & Quality Assurance: Git, Automated Unit Testing, System Monitoring, Code Reviews, Performance Profiling

EDUCATION

San Jose State University **San Jose, CA**
Master of Science in Software Engineering

ACHIEVEMENTS

- **Winner – Davis Wright Tremaine LLP Prize**, *Stanford Codex LLM x Law Hackathon (2025)*
  Led a 4-member team to build a **RAG-based legal QA engine** using Python, GPT-4, and vector-embedding retrieval with **cosine similarity**, enabling accurate extraction of contract clauses and delivering **90%+ correct answers**, outperforming 30+ teams.